information.

The motivation behind the information-theoretic interpretations was relatively simple: if the wave function is information of some sort, rather than being a physical object, then many of the puzzles at the heart of quantum physics seem to melt away. In particular, the measurement problem seems much easier to explain if the wave function is information—your information changes when you make a measurement, so it's no surprise that wave functions change dramatically when measurements occur. And the EPR experiment and Bell's theorem might become less puzzling too. When two photons with entangled polarization go flying off in opposite directions and we measure the polarization of one of them, we do instantly learn the polarization of the other one—but there's nothing mysterious or nonlocal about that, any more than there's something mysterious or nonlocal about being able to instantly infer the time in Buenos Aires by looking at a clock in Beijing. And since there's nothing nonlocal about this, there's no longer any puzzle about why entanglement can't be used for faster-than-light communication.

Except that can't quite be right, as any advocate of information-theoretic interpretations would point out. Bell's theorem explicitly states that photon polarizations can't be like clocks, nor like Bertlmann's socks. If wave functions are information rather than objects in themselves, they must be information of a rather peculiar sort. "*Whose* information?" demanded John Bell. "Information about *what?*" To resolve the measurement problem, information-theoretic interpretations had to answer these questions. The most immediate and Copenhagen-friendly answers were "my information" and "information about my observations"—but to Bell, such answers were profoundly inadequate. Placing observation at the center of physics smacked of positivism, a philosophy that Bell had entertained and rejected during his college days, concluding that it led inevitably to solipsism. Solipsism—the idea that you are the only person, and everyone and everything else is merely a hallucination of some kind in your own mind—was a problem that had haunted positivism from the start. Information-based interpretations of quantum physics ran the risk of collapsing into solipsism as well. If the information that the wave function represented was your information, what makes you so special? And how could different observers agree on the same information? How could your information